DBMS Assignment 3

State Elections SQL Queries

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SQL Queries

1. Number of Candidates per Constituency

Obtain the constituency Id, constituency Name and number of candidates contesting in that particular constituency.
I joined the Candidate and Constituency tables using constituencyId and then counted how many candidates are present in each constituency. I grouped the results by constituency ID and name to get separate counts.

Query :

```
SELECT
    cons.constituencyId,
    cons.constituencyName,
    COUNT(*) AS totalCandidates
FROM Candidate as c, Constituency as cons
WHERE c.constituencyId = cons.constituencyId
GROUP BY cons.constituencyId, cons.constituencyName;
```

2. Candidates with More Than 10 Votes

Obtain the candidate Name, candidate Id who got atleast 10 votes.
I joined the Vote and Candidate tables to associate votes with each candidate. Then I counted the number of votes per candidate and used HAVING to filter only those with more than 10 votes. I also combined their names using CONCAT.

Query :

```
SELECT
    c.candidateId,
    CONCAT(c.firstName, ' ', c.middleName, ' ', c.lastName) AS
       candidateName,
    COUNT(*) AS votes
FROM Vote as v, Candidate as c
WHERE v.candidateId = c.candidateId
GROUP BY
    c.candidateId, c.firstName, c.middleName, c.lastName
HAVING COUNT(*) > 10;
```

3. Constituency Vote Count

Obtain the constituency Id, constituency Name and number of votes that got casted in that constituency.
I joined Constituency, Candidate, and Vote tables to link each vote to its constituency. Then I counted the total votes cast in each constituency and grouped the results accordingly.

Query :

```sql
SELECT
    cons.constituencyId,
    cons.constituencyName,
    cons.totalVoters,
    COUNT(v.voteId) AS votesCasted
FROM Constituency as cons, Candidate as c, Vote as v
WHERE c.constituencyId = cons.constituencyId AND v.candidateId =
   c.candidateId
GROUP BY cons.constituencyId, cons.constituencyName, cons.
   totalVoters;
```

4. Campaign Events per Candidate

Obtain the candidate Id, candidate Name and number of campaign events he/she conducted.
I joined Candidate and CampaignEvent tables to find events conducted by each candidate. Then I counted the number of events per candidate and grouped by candidate details.

Query :

```sql
SELECT
    c.candidateId,
    CONCAT(c.firstName, ' ', c.middleName, ' ', c.lastName) AS
       candidateName,
    COUNT(e.eventId) AS numberOfEvents
FROM Candidate as c, CampaignEvent as e
WHERE c.candidateId = e.candidateId
GROUP BY
    c.candidateId,
    c.firstName, c.middleName, c.lastName
HAVING COUNT(e.eventId) >= 1;
```

5. Candidates with No Votes

Obtain the candidate Id, candidate Name who got zero votes along with constituency Id, constituency Name which they contested in. I joined Candidate with Constituency and used a LEFT JOIN with the Vote table. By checking where voteId is NULL, I identified candidates who did not receive any votes.

Query :

```sql
SELECT
    c.candidateId,
    CONCAT(c.firstName, ' ', c.middleName, ' ', c.lastName) AS
       candidateName,
    cons.constituencyId,
    cons.constituencyName
FROM Candidate c
JOIN Constituency cons
    ON c.constituencyId = cons.constituencyId
```

```
LEFT JOIN Vote v
    ON c.candidateId = v.candidateId
WHERE v.voteId IS NULL;
```

6. Officers and Their Subordinates

Obtain the ID, full name, and designation of each manager, along with the number of subordinates reporting to them.
I performed a self-join on the ElectionOfficer table to relate each officer with their subordinates. Then I counted how many officers report to each manager and grouped the results.

Query :

```
SELECT
    m.officerId AS managerId,
    CONCAT(m.firstName, ' ', m.lastName) AS managerName,
    m.designation AS managerDesignation,
    COUNT(e.officerId) AS numberOfSubordinates
FROM ElectionOfficer as m, ElectionOfficer as e
WHERE e.reportsTo = m.officerId
GROUP BY
    m.officerId, m.firstName, m.lastName, m.designation;
```

7. Winning Candidate Details

Retrieve the name of each winning candidate, the number of votes they secured, and the total number of votes cast in their constituency
I joined the Result table with Candidate to get winning candidates, and then joined with other candidates and votes in the same constituency to count total votes cast. GROUP BY was used to aggregate the values properly.

Query :

```
SELECT
    CONCAT(c.firstName, ' ', c.middleName, ' ', c.lastName) AS
       candidateName,
    r.totalVotes AS votesWon,
    COUNT(v.voteId) AS totalVotesCast,
    r.constituencyId
FROM Result as r, Candidate as c, Candidate as c2, Vote as v
WHERE r.winnerStatus = 'Winner'
  AND r.candidateId = c.candidateId
  AND c2.constituencyId = r.constituencyId
  AND v.candidateId = c2.candidateId
GROUP BY r.candidateId, r.constituencyId;
```

8. Polling Booth with Maximum Turnout

Find the polling booth(s) with the highest voter turnout ratio
I retrieved details of each polling booth including its ID, location, capacity, total votes

cast, and turnout ratio (votes cast divided by capacity). I used COUNT() with GROUP BY to calculate the total votes and turnout for each booth. I then used a subquery with MAX() to return only those booth(s) that have the highest turnout ratio among all.

Query :

```
SELECT
    pb.boothId,
    pb.location,
    pb.capacity,
    COUNT(v.voteId) AS votesCasted,
    (COUNT(v.voteId) / pb.capacity) AS turnoutRatio
FROM PollingBooth as pb, Vote as v
WHERE pb.boothId = v.boothId
GROUP BY
    pb.boothId, pb.location, pb.capacity
HAVING (COUNT(v.voteId) / pb.capacity) = (
    SELECT MAX(COUNT(v2.voteId) / pb2.capacity)
    FROM PollingBooth as pb2, Vote as v2
    WHERE pb2.boothId = v2.boothId
    GROUP BY pb2.boothId, pb2.capacity
);
```